

Lab 3: Creating EC2 Instances

Duration: 25 minutes · **Region:** us-east-2 (Ohio)

Objectives

- Create a key pair and launch an Amazon Linux instance into your VPC
- Connect to it from the browser and confirm it is using its IAM role

Exercise 3.1: Key Pair (2 min)

1. **EC2** → **Key Pairs** → **Create key pair**.
2. Name `userN-key`, type RSA, format `.pem` → **Create key pair**. The file downloads; keep it.

Exercise 3.2: Launch the Instance (10 min)

Instances → **Launch instances**:

Setting	Value
Name	<code>userN-web-1</code>
AMI	Amazon Linux 2023 (Quick Start)
Instance type	<code>t3.micro</code>
Key pair	<code>userN-key</code>
Network settings → Edit → VPC	<code>userN-vpc</code>
Subnet	<code>userN-subnet-public1-us-east-2a</code>
Auto-assign public IP	Enable
Firewall	Select existing security group → <code>userN-web-sg</code>
Advanced details → IAM instance profile	<code>userN-ec2-role</code>

Launch instance → **View all instances**. Wait for **Running** and **2/2 checks passed** (1–2 min).

Exercise 3.3: Look Around (4 min)

Select `userN-web-1` and check the tabs at the bottom:

- **Details:** Public IPv4 address, Private IPv4 address, Availability Zone, IAM Role
- **Security:** the rules from `userN-web-sg`
- **Storage:** the 8 GiB EBS root volume

Exercise 3.4: Connect (9 min)

1. **Connect** → **EC2 Instance Connect** tab → **Connect**. A terminal opens in the browser.
2. Run:

```
cat /etc/os-release | head -2      # which OS
ec2-metadata -i -z                 # instance ID and AZ
aws sts get-caller-identity        # who am I? → the userN-ec2-role
aws s3 ls                          # allowed by the role, no access keys anywhere
```

If Connect fails: check the instance is Running, has a public IP, and `userN-web-sg` allows SSH. If `aws s3 ls` fails, the IAM role was not attached.

Leave the terminal open — Lab 4 continues here.

Keep running

`userN-web-1` — used in Labs 4, 5 and 7.

Knowledge Check

1. Where did the credentials for `aws s3 ls` come from?
2. What did the AMI decide, and what did the instance type decide?